

Interpretable Joint Movement Classification via Combinatorial DTW Medoid Distances and Ensemble Feature Selection

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Abstract—Wearable inertial measurement units (IMUs) are increasingly used in physical rehabilitation to monitor upper-limb movement, yet conventional sliding-window approaches can obscure the temporal structure of individual motion cycles. Joint level recognition from a single wrist-worn sensor is particularly challenging because wrist, elbow, and shoulder movements may produce partially overlapping inertial patterns. This paper proposes an interpretable Dynamic Time Warping (DTW) template matching framework based on complete, biomechanically meaningful movement cycles rather than arbitrary temporal segments. To preserve clinical interpretability, the reference prototypes are defined as class medoids, corresponding to actual recorded participant cycles instead of synthetic average templates. For each query cycle, multivariate DTW distances are computed to these medoids over structured individual-channel, channel-pair, and channel-triplet combinations, producing an over-complete but interpretable feature representation. RUSBoost-based feature selection is then used to retain the most discriminative coordination patterns before classification with a cubic Support Vector Machine (SVM). The resulting pipeline reduces the initial 3477-dimensional combinatorial feature pool to 75 selected features. Using this compact representation, the cubic SVM achieves 91.88% overall accuracy for wrist, elbow, and shoulder movement classification, exceeding the performance of the full unpruned feature set by more than three percentage points. These results indicate that cycle-level DTW distances to clinically inspectable medoid prototypes can provide an accurate and interpretable representation for single-sensor upper-limb movement classification.

Index Terms—dynamic time warping, medoid prototype, RUSBoost, feature selection, upper limb rehabilitation, wearable IMU, support vector machine

I. INTRODUCTION

Upper limb rehabilitation relies increasingly on wearable IMUs to monitor patients and assess the quality of task practice outside clinical settings, as highlighted in recent systematic work on wearable sensor-based machine learning for human movement rehabilitation [1]. A fundamental requirement of

these monitoring systems is the ability to accurately recognise movement at the joint level, enabling practitioners to determine whether a prescribed exercise correctly engages the targeted anatomy — such as the wrist, elbow, or shoulder. Recent work on wearable rehabilitation movement classification has demonstrated the feasibility of machine learning models for distinguishing such upper-limb movement classes, while also highlighting the importance of considering deployment constraints for practical use [2].

Traditional feature extraction strategies in human activity recognition (HAR) frequently depend on hand-crafted statistical descriptors calculated over fixed sliding time windows. While computationally trivial, these approaches largely ignore the temporal trajectory of the movement and the specific multi-axis coordination that biomechanically distinguishes different joint motions. Conversely, end-to-end deep learning methods achieve high accuracy but sacrifice crucial clinical interpretability: a clinician cannot query the network to understand which sensor axes influenced the classification of an impaired movement.

Dynamic time warping (DTW) offers a natural alternative by finding optimal nonlinear temporal alignments between sequences that rarely unfold at the same pace. Early gait and activity work paired DTW with a 1-nearest-neighbour classifier — an approach difficult to surpass on standard benchmarks but limited by linear test-time cost in the training set size [3]. Kate reframed this relationship by proposing that DTW distances to a small set of reference sequences be treated as features for a separate classifier, achieving consistent improvements over 1-NN DTW across 31 of 47 datasets [3]. Subsequent work has extended the paradigm: Shokoohi-Yekta et al. show that adapting independent versus dependent warping per channel improves recognition on multi-sensor data [4]; Seo et al. use DTW similarity to assess stroke-survivor task quality at 92% accuracy [5]; and Yuan et al. demonstrate that incorporating local slope information into the alignment cost yields further gains [6]. However, prior implementations frequently rely on synthetic average prototypes that lack clinical meaning and predominantly analyse single channels, ignoring the rich

combinatorial feature space of channel pairs and triplets.

The quality of any DTW-feature pipeline depends heavily on its reference templates. Two dominant strategies exist: point-by-point averaging of aligned sequences, and iterative methods such as DTW barycentric averaging, which preserves more morphological structure. Both produce mathematical constructions that correspond to no movement a real person ever performed. Gomaa and Khamis identify prototype quality as the single largest determinant of DTW-feature generalisation [7], yet the clinical usability of the prototype itself has received limited attention. The medoid — defined as the real training cycle that minimises total DTW distance to all others in its class — retains the discriminative properties of an averaged centroid while remaining an observable, clinically meaningful movement.

On the classification side, cubic polynomial kernels reach the highest reported accuracies on wearable-sensor classification of functional movements [9], and SVMs trained on IMU features generalise well across participants [10]. Recent upper-limb IMU studies have further shown that systematic feature engineering, dimensionality reduction, and embedded feature selection can improve the discrimination of wrist, elbow, and shoulder movements [11], [12]. Boosting-based feature ranking has been validated for high-dimensional, class-imbalanced biomedical problems [13], and the two-stage paradigm of boosted selection followed by SVM classification outperforms forward selection and PCA on related tasks [14]. In a related classification setting, Ghimire and Lee demonstrated that extracting combinatorial landmark pairs and filtering them with a boosted ensemble prior to SVM classification effectively isolates discriminative patterns [15]. IMU-based literature has also explored automated feature pipelines and ensemble models for activity recognition and clinical balance assessment [16]–[18], underscoring the need to balance performance with inspectable feature representations [19].

Four gaps emerge from this literature. DTW has been validated as a feature extractor in general time-series settings but not applied to joint-level upper limb classification. Most HAR research targets coarse whole-body activities rather than identifying which joint produced a movement [7], [8]. Existing DTW-feature pipelines compute distances on single channels, leaving combinatorial subsets of axes unexplored. And reference prototypes have almost universally been mathematical averages that no clinician can directly inspect.

To address these gaps in concert, this paper introduces a clinically oriented cycle-level DTW template-matching framework for joint-level upper limb movement classification. The pipeline departs from sliding window methods by first detecting and normalising complete, observable motion cycles. Reference prototypes are strictly defined as medoids — actual recorded cycles from the training data — ensuring that system baseline behaviours can be replayed and visually inspected by therapists. The framework then computes multivariate distances across systematically structured individual, paired, and triplet channel combinations, and an ensemble-based selection mechanism passes a compact, interpretable subset to a support

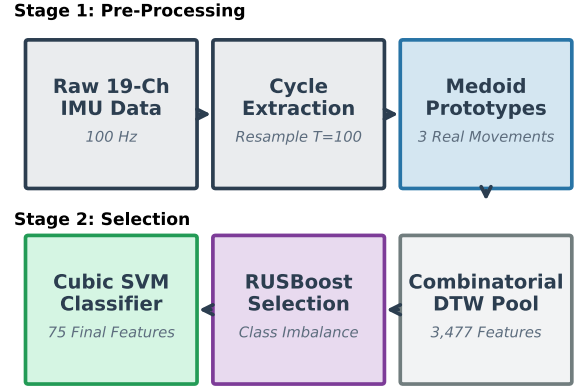


Fig. 1. End-to-end pipeline for reducing the combinatorial DTW feature pool to an interpretable subset.

vector machine for final classification.

II. METHODOLOGY

The proposed joint movement classification system processes raw IMU signals through four sequential stages: motion cycle extraction and normalisation, DTW-based feature extraction, RUSBoost-guided feature selection, and SVM classification. A visual overview of this end-to-end workflow is provided in Fig. 1.

A. Data Preprocessing and Motion Cycle Extraction

The evaluation utilises the HemiPhysioData dataset, collected via a custom WatchOS application on Apple Watch Series 6 devices. Data were recorded from three adult participants who performed three canonical upper limb rehabilitation exercises: wrist pronation/supination, elbow extension/flexion, and shoulder extension/flexion. Through Apple’s CoreMotion framework, the wrist-worn IMU captured 19 distinct data channels per recording at 100 Hz, encompassing raw acceleration, user acceleration, gravitational components, rotation rate, Euler angles (roll, pitch, yaw), and attitude quaternions (w, x, y, z) .

The raw IMU signals were segmented by subject and movement class. For each segment, the highest-variance channel served as the reference for cycle detection. Cycle length was estimated via normalised autocorrelation, with peaks searched in the 0.5–3.0 s range, consistent with the bandpass filtering strategies used for upper limb rehabilitation exercises [20]. Each accepted cycle was resampled to $T = 100$ time points and standardised channel-wise:

$$\tilde{x}_{t,c} = \frac{x_{t,c} - \mu_c}{\sigma_c}, \quad \sigma_c = \max(\hat{\sigma}_c, 10^{-8}), \quad (1)$$

where μ_c and $\hat{\sigma}_c$ are the sample mean and standard deviation of channel c over the cycle.

B. Prototype Construction

One prototype per movement class was constructed from training data exclusively. Rather than averaging aligned sequences, the class prototype was defined as the medoid — the

training cycle minimising total DTW distance to all others in its class:

$$s^* = \arg \min_{s_i \in \mathcal{C}} \sum_{j=1}^{|\mathcal{C}|} d_{\text{DTW}}(s_i, s_j), \quad (2)$$

where $\mathcal{C}$ denotes the set of training cycles for a given movement class and $d_{\text{DTW}}(\cdot, \cdot)$ is the DTW distance. Each prototype is a $T \times 19$ matrix of real recorded data, directly inspectable by a clinician.

C. DTW-Based Feature Extraction

A structured feature pool was constructed by computing DTW distances between each motion cycle and each class prototype, evaluated over systematically defined subsets of the 19 IMU channels. Drawing upon combinatorial pair and group modeling concepts [15], three types of channel subsets were defined:

- **Type 1** — Individual channels: $\binom{19}{1} = 19$ subsets
- **Type 2** — Channel pairs: $\binom{19}{2} = 171$ subsets
- **Type 3** — Channel triplets: $\binom{19}{3} = 969$ subsets

This produces 1,159 channel combinations across all three types. For each combination and each of the three class prototypes, the multivariate DTW distance was computed using [21]:

$$d_{\text{DTW}}(X, Y) = \min_W \sum_{(i,j) \in W} \|x_i - y_j\|_2, \quad (3)$$

where W is a warping path satisfying standard boundary, continuity, and monotonicity conditions. Each (combination, prototype) pair produces one scalar feature, yielding a feature vector of dimensionality $1,159 \times 3 = 3,477$ per cycle.

D. Feature Selection via RUSBoost

Predictor importance scores were derived from a trained RUSBoost ensemble, which integrates random undersampling within each boosting iteration to handle natural class imbalance arising from unequal cycle counts across subjects and sessions [22]. Features were ranked by importance and the top $N = 75$ retained, selected empirically by evaluating $N \in \{50, 75, 100, 200, 300, 400, 500\}$ on training data (Table I). Selected features were fixed prior to any test-set evaluation to prevent leakage.

The RUSBoost ensemble was configured with 30 learning cycles, a learning rate of 0.1, and decision tree weak learners, trained using MATLAB's `fitcensemble` function with random undersampling applied within each boosting iteration to address class imbalance.

E. SVM Classification

A cubic polynomial SVM was trained on the 75-dimensional reduced feature set. The cubic kernel captures nonlinear inter-feature dependencies — particularly between overlapping channel combinations — while avoiding the bandwidth sensitivity of RBF kernels. The regularisation parameter $C = 1$ was chosen via grid search over $\{0.1, 1, 10, 100\}$. For the three-class problem, a one-versus-one decomposition was

TABLE I
OVERALL AND PER-CLASS ACCURACY VS. FEATURE COUNT (* SELECTED)

N	Overall	Elbow	Shoulder	Wrist
50	91.01%	91.47%	84.35%	98.02%
75*	91.88%	90.70%	89.57%	96.04%
100	90.43%	92.25%	82.61%	97.03%
200	87.83%	87.60%	80.00%	97.03%
300	87.25%	87.60%	77.39%	98.02%
400	87.25%	90.70%	85.22%	85.15%
500	87.54%	88.37%	78.26%	97.03%
All (3,477)	88.41%	87.60%	80.00%	99.01%

TABLE II
PER-CLASS METRICS: 75-FEATURE CUBIC SVM

Class	Acc.	Prec.	Recall	F1	AUC
Elbow	90.70%	0.900	0.907	0.903	0.981
Shoulder	89.57%	0.880	0.896	0.888	0.982
Wrist	96.04%	0.990	0.960	0.975	0.998
Macro	91.88%	0.923	0.921	0.922	0.987

used with class assignment by majority vote across pairwise classifiers.

III. EXPERIMENTAL RESULTS

A. Feature Subset Selection

The Cubic SVM was evaluated across a range of values of N with all other pipeline components held fixed. As shown in Table I, accuracy peaks at $N = 75$ (91.88% overall) and declines in both directions. Using the full 3,477-dimensional feature space yields 88.41%, over three percentage points below the 75-feature model. All subsequent results use $N = 75$.

B. Classification Performance

The 75-feature Cubic SVM achieves 91.88% overall accuracy. Table II reports the full per-class breakdown. Wrist is the strongest across all measures (F1: 0.975, AUC: 0.998, AP: 0.986). Elbow and shoulder are comparable in accuracy and F1-score, but diverge in average precision: elbow AP is 0.624 against shoulder AP of 0.896.

Fig. 2 shows the confusion matrix. The dominant error pattern is bidirectional elbow–shoulder confusion — 12 elbow cycles predicted as shoulder and 11 shoulder cycles as elbow. Wrist incurs only 4 total errors across 101 cycles, with zero cross-confusion to elbow.

C. ROC and Precision–Recall Analysis

Fig. 3 shows the one-vs-rest ROC curves. All three detectors achieve high AUC (elbow: 0.981, shoulder: 0.982, wrist: 0.998). The near identical elbow and shoulder AUC values stand in contrast to the AP gap in Table II. The precision–recall curves (Fig. 4) make this gap visible: wrist and shoulder sustain high precision across nearly the full recall range, while the elbow curve degrades more steeply past moderate recall. The score distributions in Fig. 5 show why — true elbow positives and true shoulder negatives occupy overlapping regions of the elbow detector's score space.

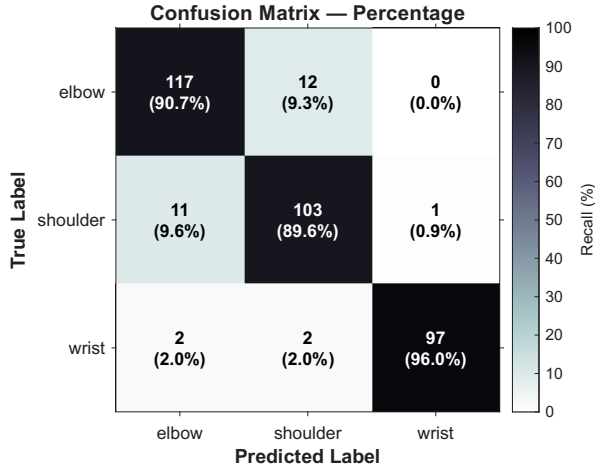


Fig. 2. Confusion matrix (% recall) for the 75-feature Cubic SVM.

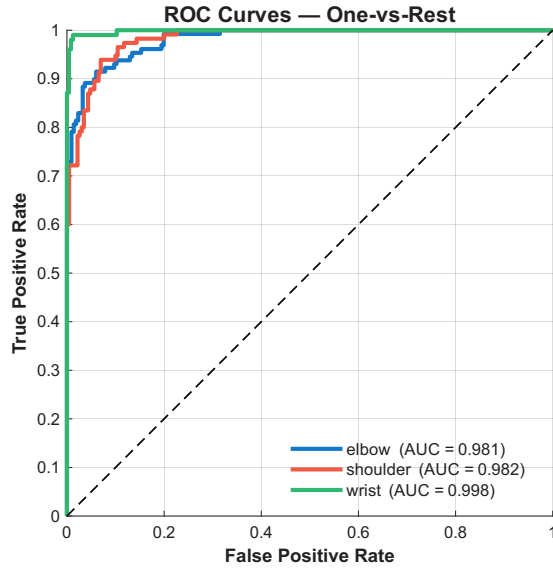


Fig. 3. One-vs-rest ROC curves for the 75-feature Cubic SVM.

D. Comparison with Baseline Methods

To contextualise the proposed pipeline against standard time-series classification benchmarks, two baselines were evaluated on the same test set: a Euclidean 1-NN classifier (no temporal alignment) and a multivariate dependent 1-NN DTW classifier, both operating directly on the resampled, standardised 19-channel cycles without feature engineering. Table III reports overall and per-class accuracy across all methods.

Euclidean 1-NN trails 1-NN DTW by 7.25 percentage points overall, with the largest deficit observed in shoulder movements (66.96% vs. 84.35%), empirically justifying non-linear temporal alignment for movements subject to inter-cycle timing variability. While the proposed 75-feature pipeline achieves the highest overall accuracy (91.88%), a McNemar's test on paired predictions indicates that the gain over 1-NN DTW is not statistically significant ($\chi^2 = 1.88$, $p = 0.170$). The primary contribution of the proposed framework over 1-NN DTW therefore rests on inference efficiency and clinical

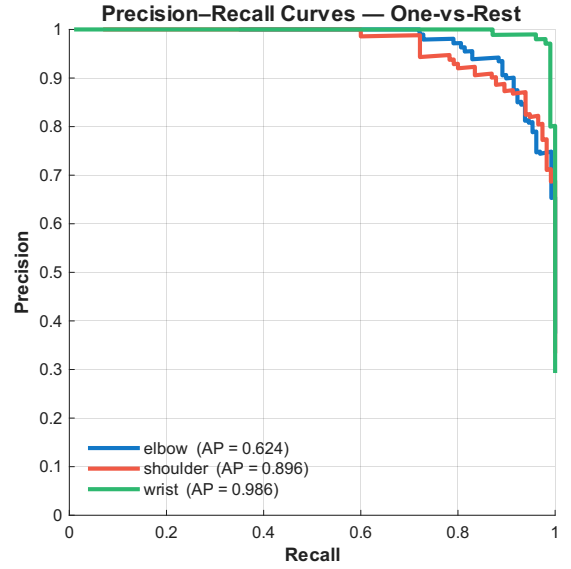


Fig. 4. Precision-recall curves for the 75-feature Cubic SVM. Elbow AP (0.624) is substantially lower than shoulder (0.896) and wrist (0.986), capturing precision degradation at high recall that AUC alone does not surface.

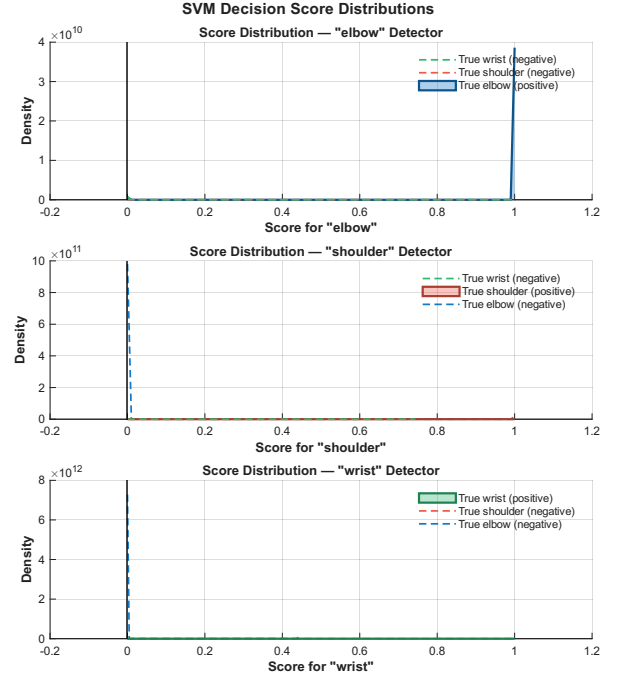


Fig. 5. SVM decision score distributions per class. True elbow positives and true shoulder negatives overlap in the elbow detector's score space, explaining the AP gap.

transparency: 1-NN DTW requires an expensive 19-channel DTW alignment against every training cycle at query time, whereas the proposed method evaluates DTW distance against only three static medoids over 75 selected low-dimensional sub-channel combinations (1–3 axes), each mapping directly to an inspectable kinematic descriptor.

IV. DISCUSSION

The observation that $N = 75$ outperforms the full feature space by over three percentage points suggests that RUSBoost

TABLE III
COMPARISON WITH BASELINE METHODS

Method	Overall	Elbow	Shoulder	Wrist
Euclidean 1-NN	82.32%	86.82%	66.96%	94.06%
1-NN DTW	89.57%	90.70%	84.35%	94.06%
Proposed (75f Cubic SVM)	91.88%	90.70%	89.57%	96.04%

ranking separates the discriminative core of the over-complete DTW space from its redundant tail. The non-monotonic accuracy curve is consistent with well-known behaviour of kernel SVMs under increasing dimensionality: beyond an effective operating point, additional features add variance rather than signal. That this point falls well below 2% of the total feature count further supports the practical utility of the ranking step.

The baseline comparison in Table III reinforces this picture from a different angle. The gap between Euclidean and DTW-based 1-NN (7.25 points) confirms that temporal alignment itself is necessary, particularly for shoulder movements where inter-cycle timing variability is greatest. The proposed pipeline’s accuracy gain over 1-NN DTW does not reach statistical significance ($\chi^2 = 1.88$, $p = 0.170$), an honest limitation of the current sample size that does not diminish the framework’s practical value: unlike 1-NN DTW, which requires an alignment against every training cycle at inference time, the proposed method requires distance computation against only three fixed medoids, and every one of its 75 retained features is independently inspectable. Feature selection is therefore not merely a compression step — it is what converts an over-complete but theoretically inspectable representation into one that is practically interpretable, while simultaneously reducing the DTW computation count roughly 46-fold relative to the full 3,477-feature pool.

The per-class performance hierarchy may partly reflect underlying biomechanics. Wrist pronation/supination produces a distinctive gyroscope signature that appears well-separated from elbow and shoulder motion in the DTW prototype space. The elbow–shoulder confusion — 12 elbow misclassified as shoulder and 11 shoulder as elbow — is consistent with the kinematic coupling commonly observed between these anatomically proximate joints.

The most operationally important finding is the elbow AP of 0.624. For this class, accuracy (90.70%) and AUC (0.981) both appear satisfactory, yet the precision–recall curve shows that precision drops sharply once the model is pushed beyond moderate recall — a trade-off that neither metric surfaces independently. Reporting AP alongside AUC changes the interpretation and is therefore not a formality in this context.

Finally, each of the 75 retained features maps directly to a specific IMU channel combination and prototype pair, so the selection outcome is fully inspectable rather than opaque — a practical property for clinical contexts where a classification decision may need to be audited or explained.

V. CONCLUSION

This paper introduced a clinically interpretable DTW medoid template-matching framework for joint-level upper

limb movement classification. By defining prototypes as medoids of real recorded data, computing DTW distances across individual, paired, and triplet channel combinations, and applying RUSBoost feature selection, the pipeline reduces 3,477 candidate features to a compact 75-feature subset that achieves 91.88% overall accuracy on a cubic SVM — outperforming the full feature space by more than three percentage points. The interpretability of both the prototypes and the selected features makes the approach well-suited for clinical deployment. In future embedded rehabilitation systems, lessons from real-time deep learning deployment on constrained devices may also inform model compression, latency-aware implementation, and practical edge-AI integration [23].

This evaluation is based on a three-participant dataset, and the accuracy gain over the 1-NN DTW baseline, while consistent (91.88% vs. 89.57%), did not reach statistical significance under a paired McNemar’s test ($\chi^2 = 1.88$, $p = 0.170$) — limitations that should be addressed through larger, more diverse cohorts. Future work should validate the framework under subject-independent cross-validation, explore multi-joint prototype representations to reduce elbow–shoulder overlap, and formally benchmark inference latency against 1-NN DTW to quantify the deployment advantage suggested in this study.

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